

deConfUSlon fUSI Studio

User manual and processing reference

Version 2026-09-06

MATLAB 2017b-2023b compatible

Functional ultrasound loading, quality control, preprocessing, registration, SCM, video, connectivity, and group analysis.

Start here

Load a dataset, confirm its dimensions and timing, run Full QC, then apply only the preprocessing justified by the experiment. Every derived file keeps the source identity and appears in the dataset dropdown.

Safety of interpretation

This is research software. Review raw movies, QC, baseline windows, masks, registration overlays, and exported statistics before using results.

1. Installation and first launch

Place the repository at D:/Github/deConfUSIon, start MATLAB, add the repository to the path, and run `run_fusi_studio`. The launcher assembles the split GUI source files into a temporary runtime file and keeps backup folders off the MATLAB path.

```
cd('D:/Github/deConfUSIon')  
run_fusi_studio
```

2. Load data and confirm timing

Supported image arrays are [Y X T] for 2D and [Y X Z T] for 3D or step-motor data. The loader retains acquired frames, stores physical t sec when available, detects animal and scan identity, and records the timing source.

The Load dataset options window remains a normal centered dialog. Custom TR is selected by default and prefilled from a file timing candidate when available. Closing that dialog after the image is read continues with the automatic output folder instead of discarding the loaded dataset.

2a. What fUSI, TR, and resolution mean

Functional ultrasound imaging (fUSI) measures blood-volume or blood-flow-related changes with ultrafast ultrasound. A voxel or pixel is a spatial sample; the final dimension is the time series. A bright change is not automatically neuronal activity: motion, probe pressure, vessels, anesthesia, physiology, and acquisition drift can all change the signal. Every map therefore needs raw-movie and QC review.

Temporal resolution and TR. The repetition time (TR) is the time between consecutive acquired frames, in seconds. The nominal sampling frequency is $f_s = 1/TR$ and the Nyquist frequency is $f_N = 1/(2*TR)$. The effective TR is computed from the physical timestamp vector when one exists; an irregular timestamp sequence should be resampled explicitly before filtering or frequency-domain FC. A custom TR is entered in milliseconds in the load dialog and stored as seconds in the dataset metadata.

Spatial resolution. Pixel or voxel spacing describes the physical field of view represented by each sample. It is independent of temporal resolution. Registration, masks, atlas labels, and underlays must agree in orientation, dimensions, spacing, and slice order. A visually plausible overlay can still be wrong if the geometry is mismatched.

3D and step-motor data. A true volume is [Y X Z T]. Step-motor acquisitions may be a sequence of 2D planes reconstructed into Z; treat each source plane and frame grouping explicitly and inspect slice order before atlas or group analysis.

2b. Core mathematical conventions

Baseline and PSC. For voxel v , let $B(v)$ be the finite positive mean of selected baseline frames. The percent signal change is $PSC(v, t) = 100 * (I(v, t) - B(v)) / B(v)$. Invalid or non-positive baselines are excluded from maps and ROI averages. The baseline window is always recorded in seconds and in frame indices.

Rebaselining. If an already-normalized trace uses baseline B1 and a new baseline B2, the displayed value is transformed as $PSC2 = 100 * (PSC1 - B2) / (100 + B2)$. SCM and exported ROI values use the same transformation so that the plot and file agree.

Temporal filtering. A Butterworth filter is applied along time. Low-pass retains frequencies below FcHigh, high-pass retains frequencies above FcLow, and band-pass retains the interval [FcLow,FcHigh]. The filter order controls roll-off; cutoffs must be below the Nyquist frequency. Filtering changes temporal content and should be justified by the experiment rather than used as a generic cleanup.

Temporal interpolation. Sliding smoothing preserves the number of frames. Block subsampling reduces the number of frames and changes the effective TR to $TR_{new} = TR * block_size_frames$, where $block_size_frames = \max(1, \text{round}(\text{window_seconds}/TR))$.

QC metrics. Temporal SNR is $\text{mean}(I, t) / \text{std}(I, t)$. DVARS is the frame-to-frame root-mean-square change; global-signal scrubbing thresholds a global trace. For statistics, Pearson correlations are often transformed using $z = \text{atanh}(r)$ before group averaging, and Benjamini-Hochberg controls the false discovery rate over tested hypotheses.

3. Dataset names and saves

Derived files receive a readable processing suffix and a date stamp in the dropdown. Large 3D outputs use a temporary MAT file and a cooperative save queue: metadata is staged first, image chunks are appended, then the complete file is atomically published. A queued or saving label means finalization is still in progress.

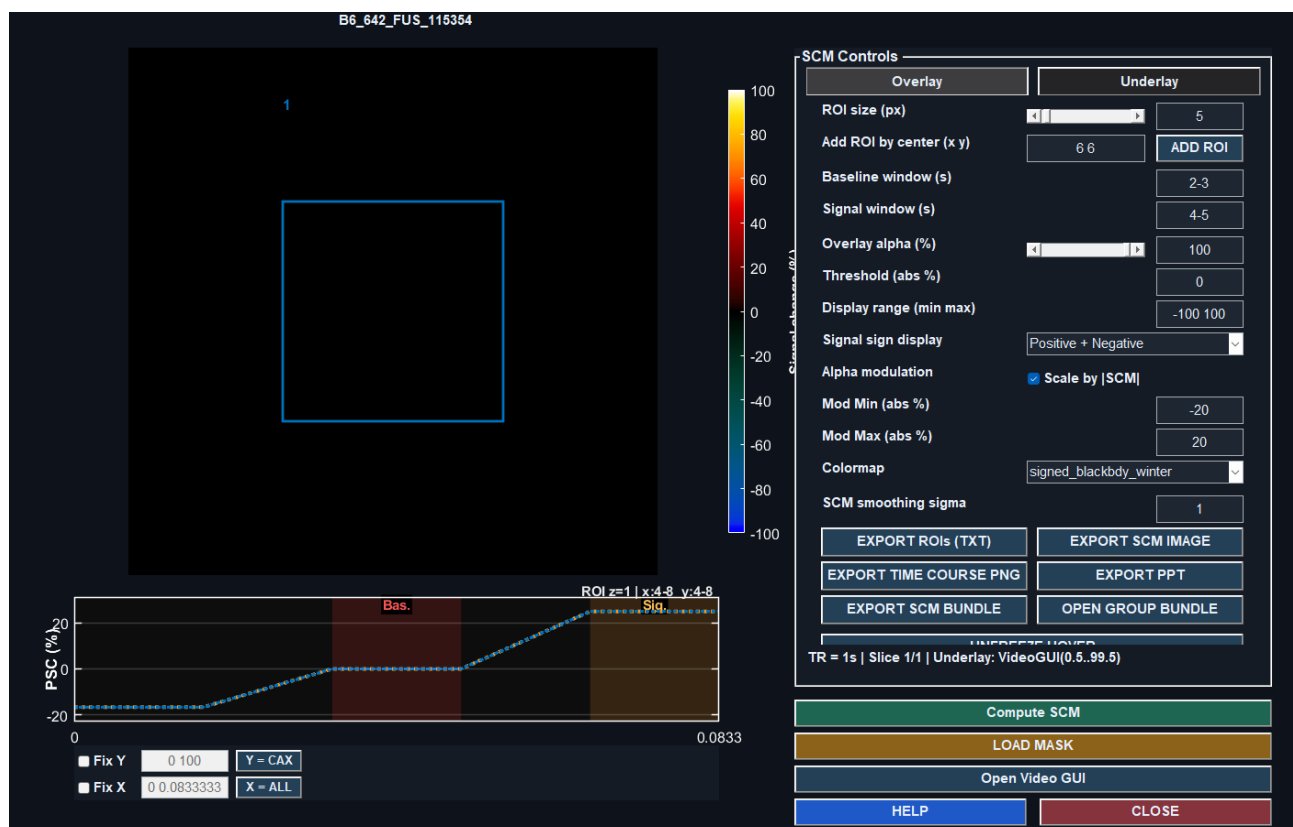
4. Motion correction and Chop Data

Motion Correction groups frame rejection, despiking, and scrubbing behind one chooser. Select a method, adjust its parameters, and press START. The raw dataset remains unchanged; the result receives a method-specific name and is immediately added to the active dataset list while its MAT file is saved in the background.

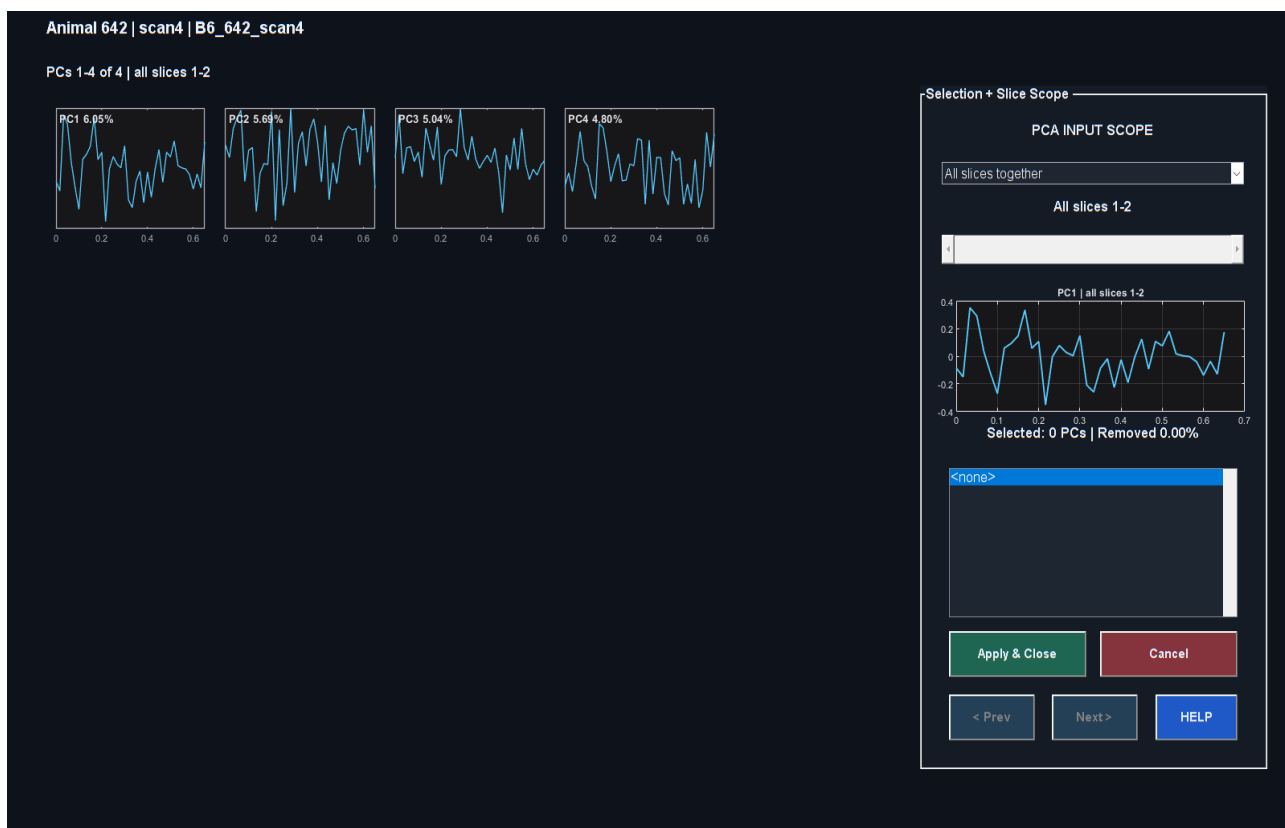
Method	Purpose	Controls
Frame rejection	Whole-frame intensity excursions; interpolate rejected volumes.	Sigma threshold; both, high-only, or low-only.
Despiking	Voxel-wise robust MAD outliers; interpolate isolated spikes.	Robust Z threshold.
Scrubbing	Frame-to-frame changes; interpolate flagged volumes.	DVARS or global signal; linear or PCHIP; optional edge trim.
Chop Data	Remove early and late physical seconds; keep original frame indices.	Start and end seconds.

5. SCM and video

SCM uses the selected baseline and signal windows in seconds. $PSC = 100 * (signal - baseline) / baseline$; SCM rebaseline is $100 * (PSC - newBaseline) / (100 + newBaseline)$. Baseline and signal edits update the map and ROI traces directly. Display range and alpha controls update without recomputing the map.



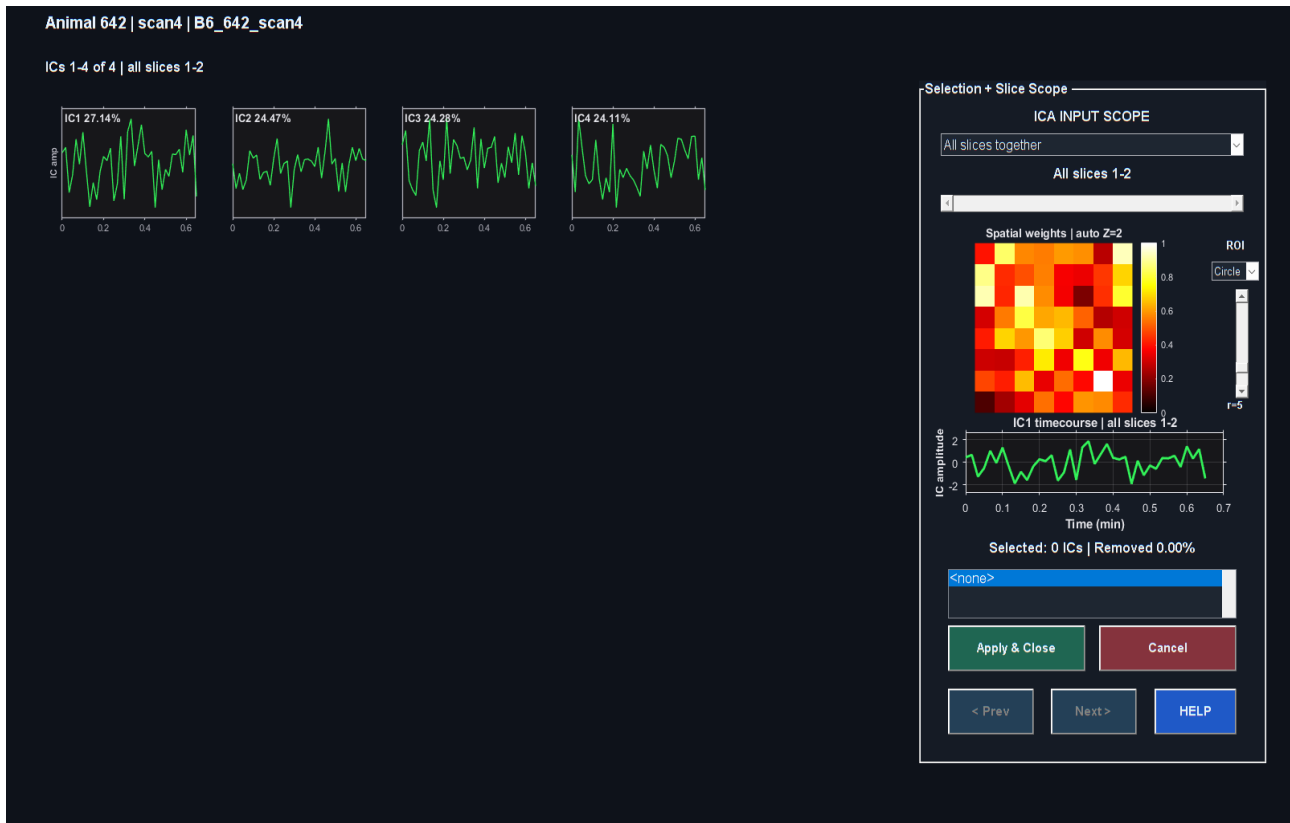
SCM GUI validation view with baseline and signal windows, ROI trace, signed color map, exports, and action controls.



PCA component review window with dataset identity and action controls.

6. PCA, ICA, filtering, and temporal interpolation

PCA and ICA are optional decomposition tools. Review components before applying removal. Filtering and Temporal Interpolation operate on the recorded timing grid and preserve the last dimension as time. Use each Help button for purpose, equations, assumptions, and output naming.



ICA component review window and validation capture.

6a. Why each preprocessing module exists

Frame rejection removes an entire frame when a global excursion makes that frame unusable. Scrubbing identifies high-change frames and interpolates them while preserving the timeline. Despiking targets sparse voxel-wise outliers while retaining a usable frame. Imregdemons corrects slow spatial deformation and must be visually reviewed because an incorrect warp can move anatomy. Drift compensation estimates a slow trend using anchor, baseline, robust, polynomial, vehicle, or PACAP-aware methods; slow biological responses can overlap drift, so the method must match the experiment.

Use the least aggressive operation that solves the QC problem. Do not automatically stack frame rejection, scrubbing, and despiking. Save each derived output with its processing chain and re-run QC after a material correction.

7. Registration, atlas regions, and segmentation

3D registration offers automatic initialization followed by manual review. Inspect several slices and orientations before saving a transform. Atlas colors are taken from the atlas label table; the preparation helper also stores a coarse anatomical grouping map for larger parent regions while retaining detailed labels for later analyses.

For the 3D example, use the true volume workflow and verify the orientation, voxel spacing, slice order, and underlay before accepting an automatic transform. For 2D or step-motor data, use the coronal workflow and inspect every slice. Coarse groups such as cortex, striatum, thalamus, and hippocampus are optional display groups; detailed labels remain available for segmentation and statistics.

8. Connectivity and group analysis

Functional connectivity reports valid sample counts and FDR-adjusted q-values. Group analysis aligns timing and subject identity before averaging. Plateau, robust peak, time-course display, and export bundles use the same physical time axis and preserve source metadata.

7a. Registration, segmentation, FC, and group reasoning

Registration estimates a transform from functional or anatomical space into an atlas space. The transform is only useful when the underlay and functional image share geometry. Segmentation samples labels from the registered atlas and extracts region time courses; inspect label boundaries and colors before interpreting a region. Coarse anatomical groups are a display and aggregation choice, not a replacement for the detailed atlas labels.

Functional connectivity computes relationships between ROI or seed time courses. Check finite sample counts, mask coverage, nuisance strategy, and the sign and scale of the matrix. Group Analysis must align subjects by identity, condition, slice, and physical time rather than blindly averaging unequal frame indices. Plateau and robust-peak summaries should be checked against the underlying time course and baseline window.

9. Mathematical conventions

Baseline normalization: $B(v)$ is the finite positive mean over baseline frames;

$PSC(v,t)=100*(I(v,t)-B(v))/B(v)$.

Rebaselining: $PSC_{new}=100*(PSC_{old}-B_{new})/(100+B_{new})$.

CompCor: nuisance components are derived from selected noise or low-variance voxels after masking and temporal residualization.

FDR: Benjamini-Hochberg correction is applied over tested hypotheses.

SVD clutter filtering: temporal singular modes are estimated from a voxel-by-time matrix and selected modes are removed according to the configured cutoff.

10. Literature and implementation references

- Demené C, et al. (2014). Functional ultrasound imaging of brain activity in rodents. Nature Methods.
- Behzadi Y, Restom K, Liao J, Liu TT. (2007). A component based noise correction method (CompCor) for BOLD and perfusion based fMRI. NeuroImage 37:90-101. DOI 10.1016/j.neuroimage.2007.04.042.
- Wang Q, et al. (2020). The Allen Mouse Brain Common Coordinate Framework: A 3D Reference Atlas. Cell 181:936-953.e20. DOI 10.1016/j.cell.2020.04.007.
- Benjamini Y, Hochberg Y. (1995). Controlling the false discovery rate. JRSS Series B 57:289-300.
- Friston KJ, et al. (1995). Statistical parametric maps in functional imaging. Human Brain Mapping 2:189-210.

11. Troubleshooting

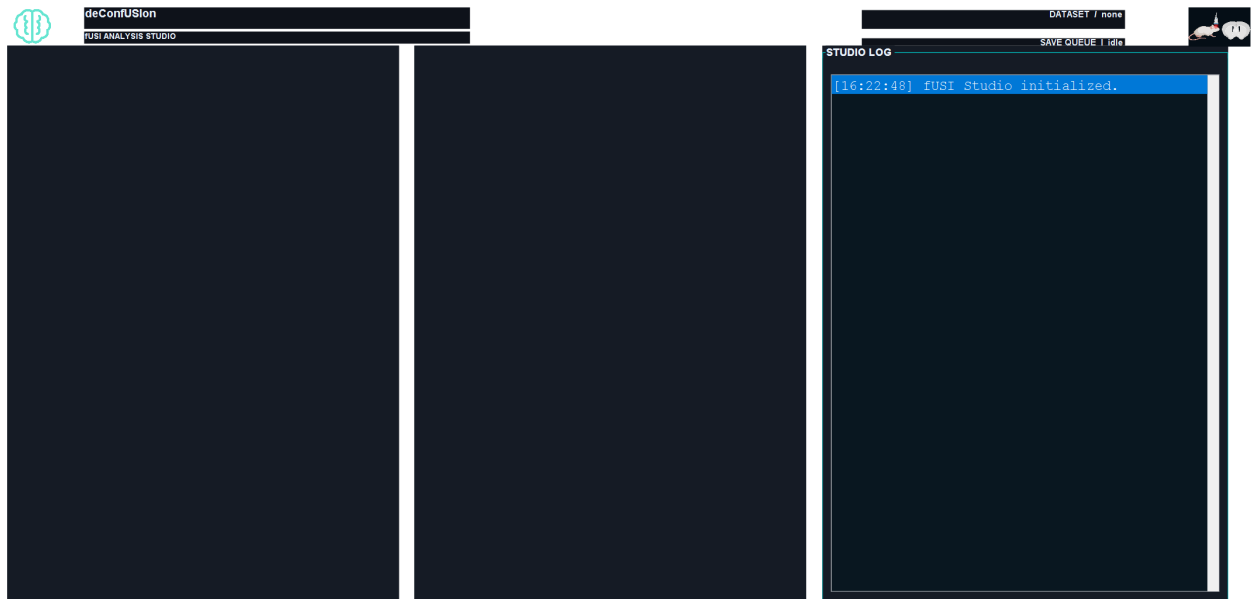
If a popup is clipped, maximize the owning window. If a dataset appears with a queued date label, wait for the save queue or open Save queue to inspect failures. If baseline values are invalid, choose finite positive baseline samples. If registration looks wrong, stop before saving and review orientation, voxel size, landmarks, and atlas plane order.

Run `validate_gui_followup`, `validate_deconfusion`, and `validate_motion_dataio` after changes.

12. Module map and figure guide

The older 24 June manual is retained as the organizing reference for this edition: purpose and scope, abbreviations, mathematical principles, installation, data organization, Studio map, standardized workflows, QC, preprocessing, visualization, masks, registration, segmentation, FC, group analysis, outputs, validation, troubleshooting, and implementation inventory. The current edition adds the repaired Studio controls, motion chooser, background saves, dated labels, coarse atlas grouping, and the 3D workflow notes above.

Figure 0. Studio header — dataset identity and the non-interactive background save-queue indicator are separated so they cannot overlap.



Module	What to inspect before running
Load fUSI Data	Dimensions, TR, time axis, source identity, output folder
Full QC	Frame outliers, tSNR, drift, baseline coverage, raw movie
Motion correction	Choose one method; confirm threshold and output name
Temporal Interpolation	Window, interpolation mode, effective TR, length
PCA / ICA	Component traces and maps; review before removal
Filtering / Drift	Cutoffs, order, sampling interval, residual mean
SCM / Video / Time Course	Baseline, signal window, range, polarity, ROI trace
Mask / Atlas / Segmentation	Underlay geometry, label colors, slice order
FC / Group Analysis	Subject IDs, timing, masks, labels, valid counts

Figure 1. SCM GUI — baseline and signal windows, map range, ROI time course, and colored actions.

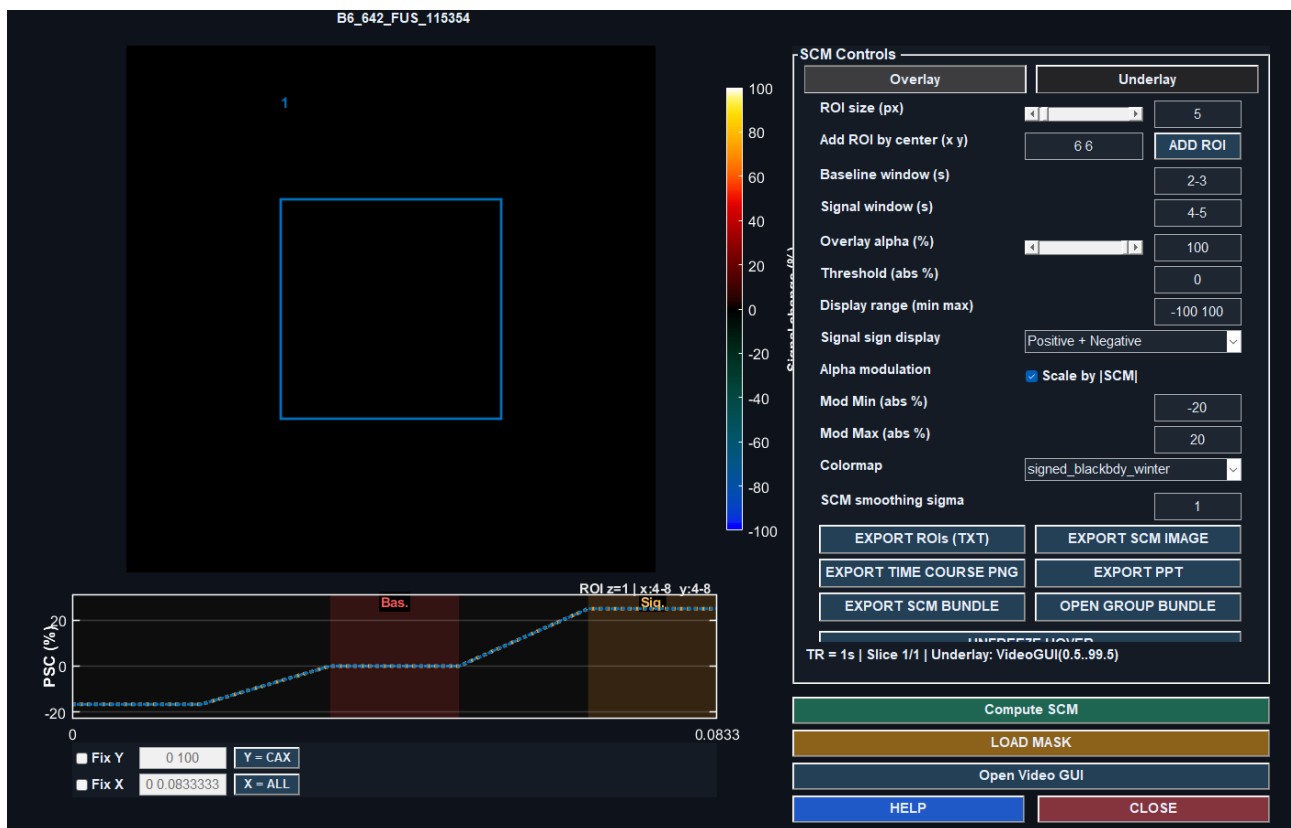


Figure 2. PCA review window — component selection and dataset identity.

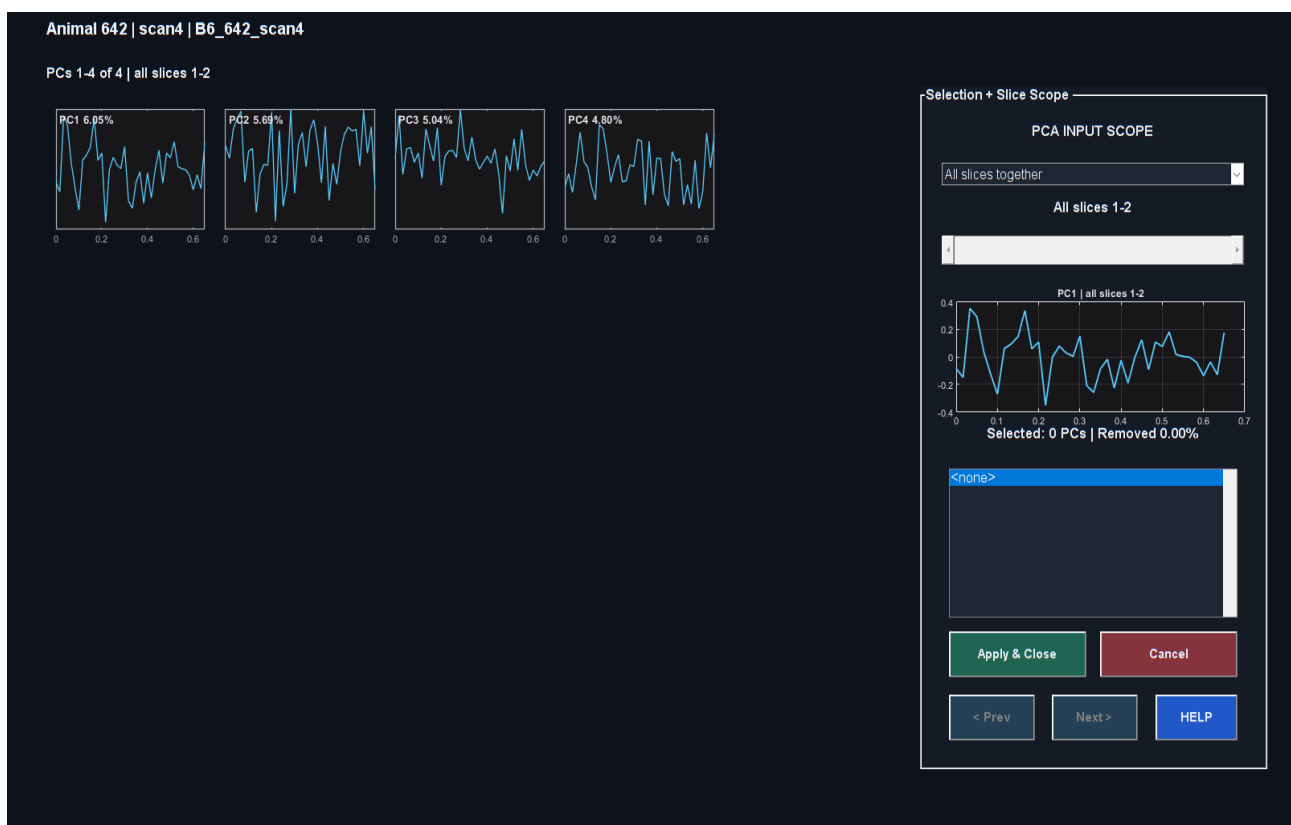


Figure 3. ICA review window — inspect independent components before applying removal.

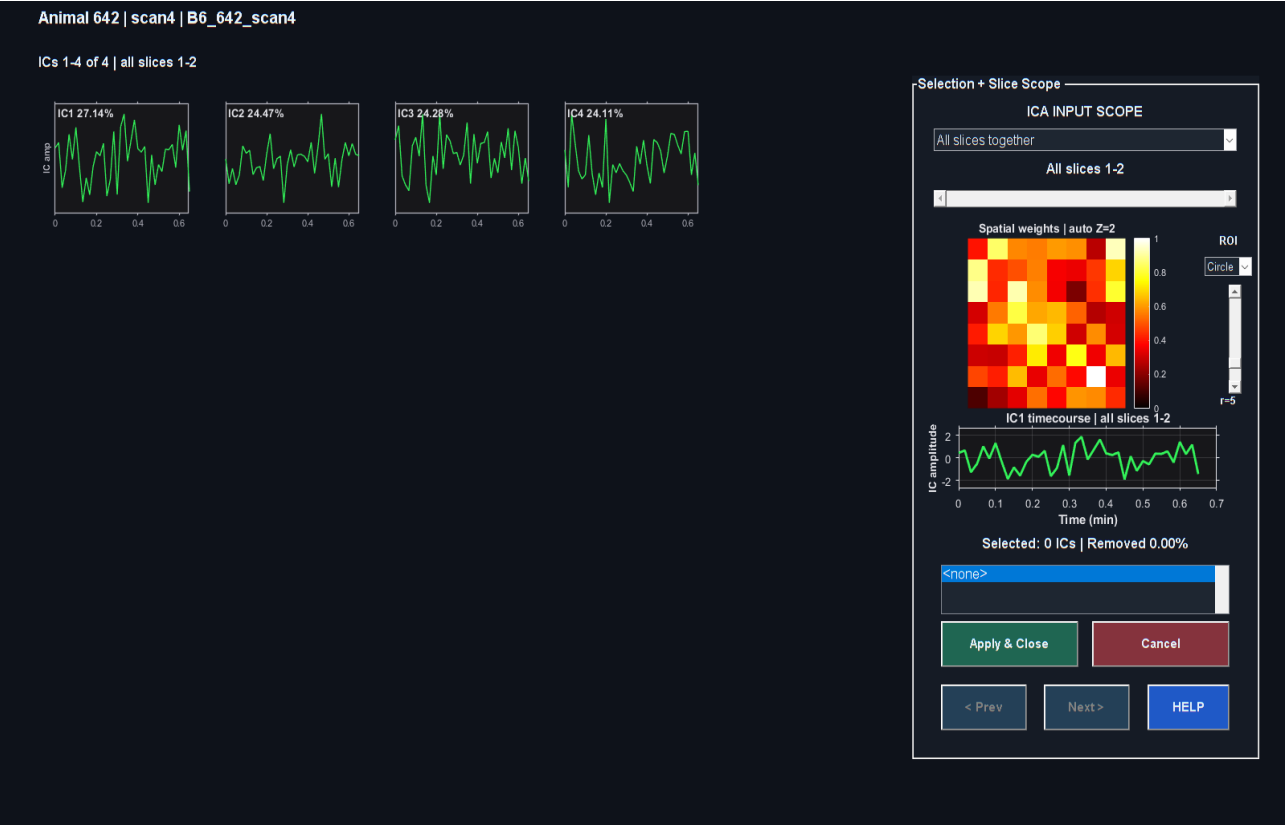
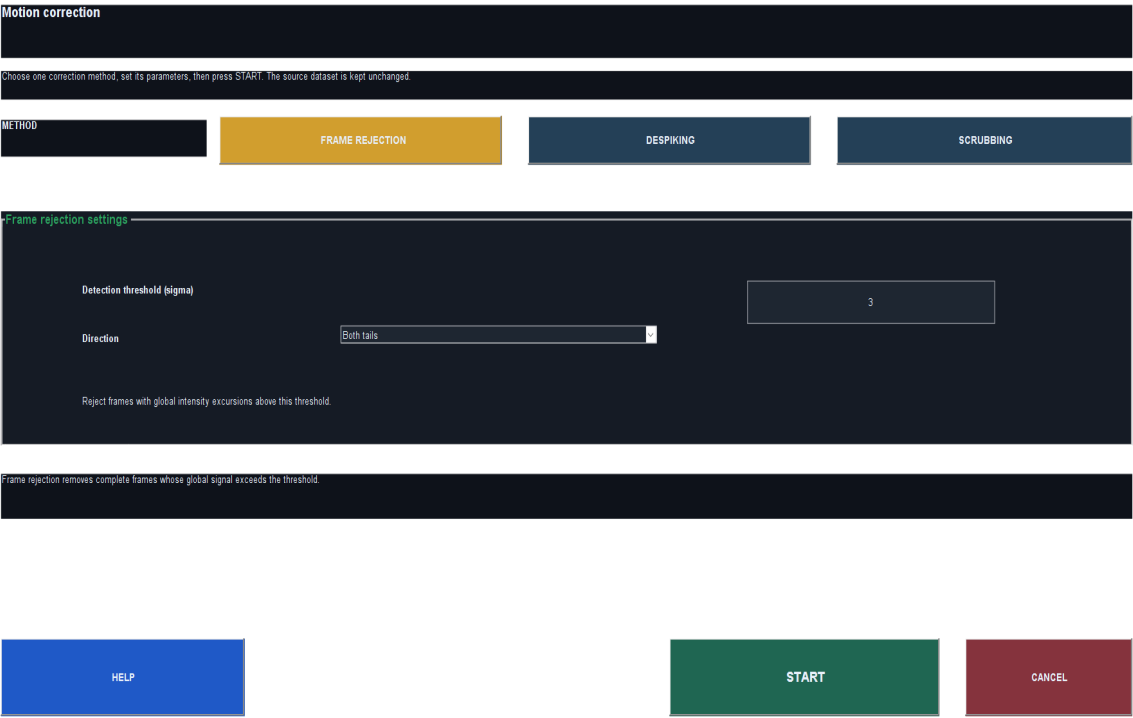


Figure 4. Motion Correction chooser — the method-specific layout keeps frame rejection, despiking, and scrubbing settings separated and aligned.



For modules without a capture in this repository edition, use the module map and the Help button inside the live GUI. Each Help topic states purpose, implementation, mathematical

convention, assumptions, and usage. Capture the live figure after loading the intended dataset when producing a project-specific report.

13. Worked example and reproducibility

Recommended example dataset (read-only): Z:\fUS\Project_PACAP_AVATAR_SC\RawData\MPI_Data\RGR0_251119_1024_MM_B6_642\RGR0_251119_1024_MM_B6_642_FUS_115354.mat. Load it, select Custom TR, verify the detected timing, run Full QC, and save only derived outputs under AnalysedData. Never edit or overwrite the raw file.

Record the dataset label with its trailing dd/mm/yy creation date, the active preprocessing chain, baseline and signal windows, mask/atlas files, and the Studio Log. This makes it possible to reproduce SCM, plateau, robust peak, time-course, FC, and group results after the background save queue finishes.

14. Change log

2026-09-06: restored colored action controls, corrected the overlapping queue indicator, made readiness state fully red/green, aligned Motion Correction panels, added Custom TR defaulting, dated labels, cooperative 3D saves, and updated this PDF from the June manual structure.